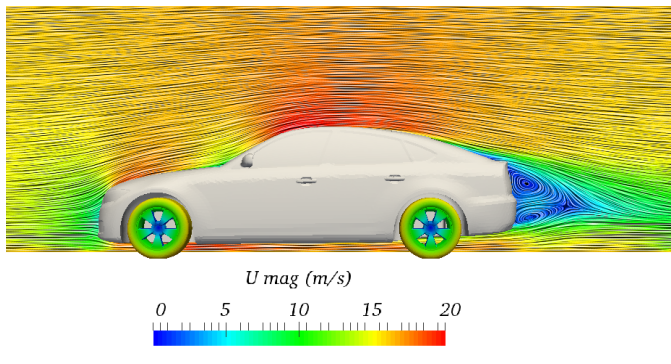


Lecture Unit 9.1

Linear algebra in HPC

Linear algebra

- Solving differential equations with a computer requires approximating calculus with algebra
- In week 3 we saw how to solve PDEs using finite differences with a forward Euler time step
- Many scientific and engineering applications use more advanced numerical methods which require solving systems of linear equations

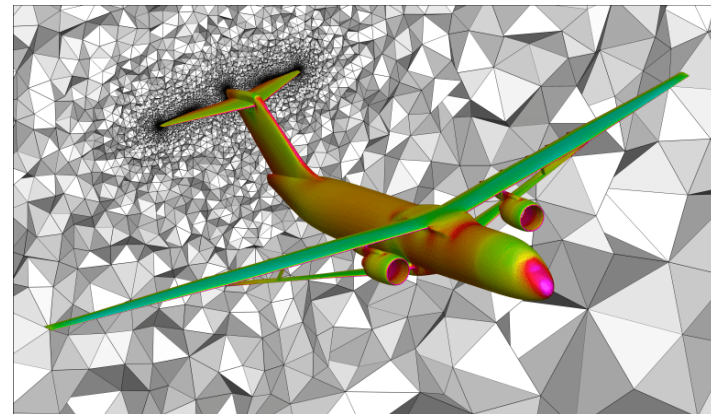
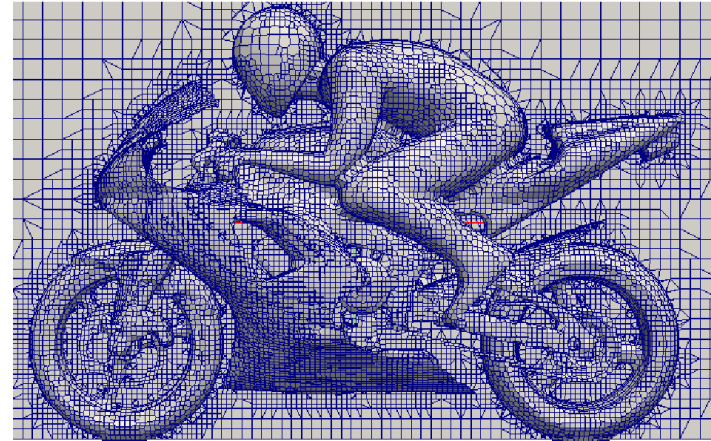


<http://emps.exeter.ac.uk/engineering/research/ourresearch/computational-engineering/>

Finite volume methods

- Finite differences are easy to implement but are not suited to complex meshes
- The finite volume method considers fluxes (e.g. the flow of mass, momentum and energy) between adjacent cells
- An example is the OpenFOAM open source, finite volume CFD software

<https://www.cfdsupport.com/OpenFOAM-Training-by-CFD-Support/node140.html>

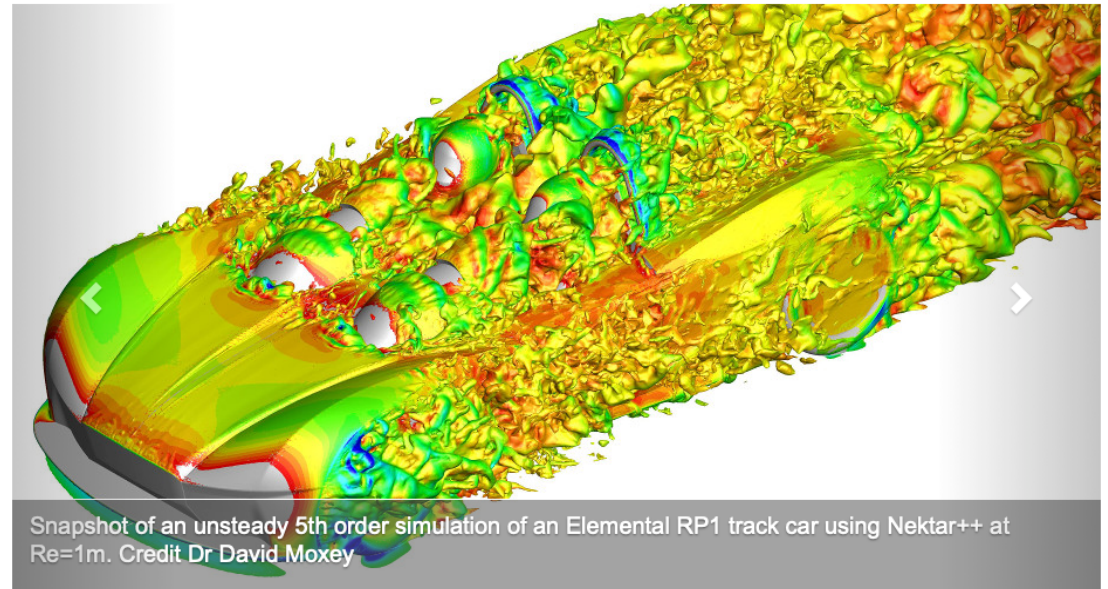


<https://blog.pointwise.com/2021/03/05/this-week-in-cfd-423/>

Finite element method

- The finite element method approximates the solution using expansion functions which are non-zero over a small region of the domain
- Two examples are Nektar++ and Firedrake
- Both finite element and finite volume methods require solving systems of linear equations to advance the solution to the next time step

Computational Engineering



Computational Engineering is the use of computers to simulate and understand engineering problems. In the Computational Engineering group we develop and use a range of techniques - Computational Fluid Dynamics, Finite Element Analysis, Surrogate Modelling, Optimisation and Machine Learning - and apply these to real world problems in industry and medicine.

Our work is supported by a range of research grants and we often work with industrial partners - take a look at some of the collaborations and projects currently active.

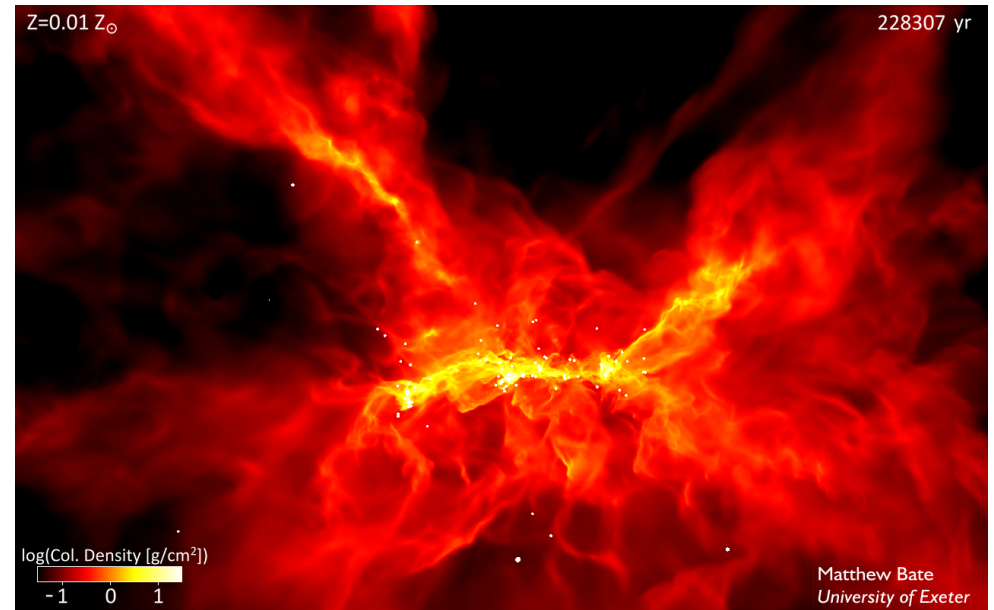
<http://emps.exeter.ac.uk/engineering/research/ourresearch/computational-engineering/>

Elliptic equations

- Some PDEs do not contain a time dependence so can't be solved with time stepping
- One example is the Poisson equation used to calculate self-gravity:

$$\nabla^2 \varphi(r) = 4\pi G \rho(r)$$

where $\varphi(r)$ is the gravitational potential, $\rho(r)$ is the density distribution and G is a constant



<http://www.astro.ex.ac.uk/people/mbate/>

Unit summary

- Solving differential equations with a computer requires approximating calculus with algebra
- To do this many scientific and engineering applications use numerical methods which require solving systems of linear equations
- Finite volume and finite element methods are two examples of methods used in computational fluid dynamics which require calculating solutions to linear systems
- Solving other types of equations (e.g. a Poisson equation) can also require solving linear systems